AMTI (NMTC) - 2004

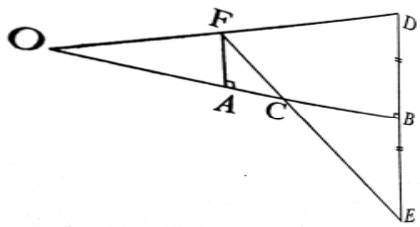
RAMANUJAN CONTEST - FINAL - INTER LEVEL

1. Find all integers n $\geq$ 1 such that $\frac{n^3+3}{n^2+7}$ is an integer.

2. A point is chosen on each side of a unit square. The four points form the sides of a quadrilateral with sides of lengths a, b, c, d. Show that $2 \leq a^2+b^2+c^2+d^2 \leq 4$
$2\sqrt{2} \leq a+b+c+d \leq 4$

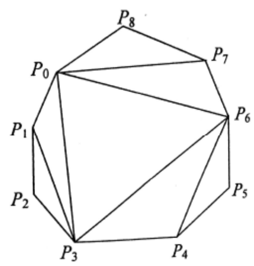
3. ABCD is a convex quadrilateral inscribed in a circle Σ. Assume that A, B and Σ are fixed and C, D are variable points, so that the length of the segment CD remains constant. Points X and Y are on the rays AC and BC respectively such that AX = AD and BY = BD. Prove that the distance between X and Y remains constant.

4. In the adjoining figure, OB is the perpendicular bisector of DE. A is a point on OB; AF is perpendicular to OB and EF intersects OB at C. Show that OC is the harmonic mean between OA and OB. i.e., $OC = \frac{2 \cdot OA \cdot OB}{OA+OB}$.



5. Find all integral values of x, y, z, w given that x! + y! = 2^z3^w.

6. A convex polygon of nine vertices P_0, P_1, P_2, ..., P_8 is given along with six diagonals as shown in diagram. We see that 7 triangles $P_0P_1P_3$, $P_0P_3P_6$, $P_0P_6P_7$, $P_0P_7P_8$, $P_1P_2P_3$, $P_3P_4P_6$ and $P_4P_5P_6$ are created. These triangles are to be numbered Δ_1, Δ_2, Δ_3, ..., Δ_7 so that P_i is a vertex of Δ_i. In how many ways can this be done ? Justify your answer.



7. Let f(x) be a linear function such that f(0) = –5 and f(f(0)) = –15. Find all values of m for which the solutions of the inequality f(x) f(m – x) > 0 form an interval of length 2.

8. In how many ways can you select two disjoint subsets from a set having n elements ?
